

LEIS Evidence-Bounded Whitepaper v1.0 - English

LEIS MEMORY

Evidence-Bounded Orientation Preservation and Reconstruction Protocol

Release: v1.0 (gatekeeper revision, 2026-08-16)

Working language: English

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Abstract

LEIS MEMORY is a method for preserving enough orientation to recover understanding after context has been compressed or transferred. It is not a truth engine, a consciousness transfer, a general intelligence certificate, or a replacement for source material. Its operational promise is narrower and testable: a bounded receiver should be able to reconstruct a declared goal, risks, decisions, lineage and unknowns from a Reality Frame, while keeping evidence boundaries visible.

The supplied predecessor, Vedecky whitepaper: LEIS MEMORY v1.0-Stabilized (2026-08-16), contains the right research direction but presents several hypotheses and local demonstrations as settled theorems or completed empirical benchmarks. This revision preserves the useful architecture and removes unsupported certainty.

1. Scope and non-claims

LEIS covers evidence-bounded reconstruction and its governance. It does not establish that:

- all meaning can be reconstructed from zero state;
- $R(U) = T L S$ is a physical theorem or a refutation of Shannon information theory;
- a high Orientation Preservation (OP) score is truth, correctness, intelligence or legal compliance;
- a local synthetic fixture generalizes to all books, domains, models or people;
- an ACCEPTEDOPERATIONAL handshake certifies a person, model or claim as infallible;
- a protocol document alone satisfies the EU AI Act, GDPR or any other law.

The phrase Reality $\neq$ Render is mandatory. A render may be observed, simulated, inferred or unknown and must be labelled accordingly.

2. Frozen operational core

The current LEIS MEMORY Core freezes the following working objects and gates:

1. Reality Frame - a bounded representation of goal, problem, risks, decisions, evidence boundary, unknowns and lineage.
2. LEISZERO - a transport/reconstruction hypothesis and contract. It is not presented here as a physical theorem.
3. GF-1 Reality Frame Reconstruction - the receiver recovers the declared orientation.
4. GF-2 Validation Agreement - an independent evaluator agrees with the validation conclusion within the declared rubric.
5. GF-3 Lineage Delta Agreement - material source or interpretation changes are recovered.
6. GF-4 Unknown Preservation - unknowns remain unknown unless new evidence promotes them.
7. OP (Orientation Preservation) - a versioned operational metric, not a truth metric.
8. Failure Taxonomy and Boundary Mapping - failures are recorded rather than silently promoted to success.

The working loop is:

Reality -> Compression -> Transfer -> Reconstruction -> Reality Check -> Lineage

3. Evidence model

Every material statement should carry, where available:

- source identifier and source class;

- version or date provenance;
- locator (page, section, span, or event identifier);
- declared confidence boundary;
- explicit unknowns and conflicts;
- the safe next action or validation requested.

The permitted states are SUPPORTED, CONFLICTED, UNKNOWN, and REJECTED. Repetition never upgrades a state. Promotion requires new evidence or a documented human decision.

4. Reconstruction protocol

The minimum experimental sequence is:

Source -> Reality Frame -> LEISZERO -> Independent Reconstruction -> GF Validation -> OP -> Review
-> Lineage

The sender retains the reference (U0) and the receiver is not given the answer key. A receipt is evaluated against the declared rubric. If a critical element is missing, the protocol emits a bounded PATCH or returns HOLD; it does not silently fill gaps.

The protocol is suitable for human, AI and mixed observers. It measures recoverability of orientation, not inner experience or semantic identity.

5. OP metric and calibration boundary

The current pilot definition is the mean of four recovery dimensions:

OP = mean(Goal Recovery, Risk Recovery, Decision Recovery, Lineage Recovery)

Each dimension is scored against a predeclared rubric. Unknown preservation is a separate gate and cannot be hidden inside a favourable average. Thresholds are study parameters. A score must always be reported with its fixture, rubric, observers, sample size, source boundary and confidence interval or explicit absence of calibration.

Existing local evidence includes synthetic Gate-01 and Understanding Stream fixtures and one real German spatial-planning reconstruction (reported OP approximately 0.89). These are useful demonstrations, not proof of universal performance. The random multi-domain intake remains an active research programme; the exact eight-domain scores stated in the predecessor paper are not accepted as verified evidence in this release.

6. Findings from the gatekeeper audit

Retained

- orientation preservation is a meaningful engineering target;
- Reality Frames, unknown preservation, provenance and lineage deltas are useful controls;
- Socratic challenge/response is a practical way to test reconstruction without handing over the answer key;
- fail-closed handling of conflicts and missing anchors is preferable to confident completion;
- Memory should return bounded evidence atoms and locators, not entire private books or transcripts.

Reclassified as candidate or research

- the LEISZERO equation and entropy notation;
- claims of complete reconstruction from absolute zero;
- decay-plateau simulations as evidence of general cognition;
- the full Memory Graph ontology and public query brain;
- exact cross-domain OP values without reproducible fixtures and independent review;
- claims of compliance or certification under any law.

Removed from the scientific claim set

- any assertion that LEIS refutes Shannon's theory;
- any implication that a local test proves truth, consciousness, or universal intelligence;
- any automatic legal conclusion from the presence of a schema or audit trail.

7. Validation roadmap

The next evidence required is deliberately small and falsifiable:

1. Run the same blinded fixture across unrelated domains selected by a predeclared random procedure.
2. Use at least one human reality observer and one independent boundary observer.
3. Keep the source and U0 reference private to the evaluator.
4. Record GF-1 through GF-4, OP components, unknowns, conflicts, and lineage deltas.
5. Publish fixtures, rubrics, raw receipts and failure cases, not only aggregate scores.
6. Repeat until the study can show both preservation boundaries and failure boundaries.

The research question is not “Can LEIS reconstruct this book?” but “Which kinds of orientation survive when most information is gone, and where does the method fail?”

8. Legal and governance boundary

The EU AI Act is a source of design considerations, not an automatic approval. Article 12 concerns logging and traceability for high-risk systems; Article 14 concerns human oversight. Article 99 sets penalty ceilings, including a lower SME ceiling for certain incorrect, incomplete or misleading information, but applicability depends on role, conduct, jurisdiction and the current law. No LEIS document is a legal opinion or conformity assessment.

Primary source: [Regulation (EU) 2024/1689 on EUR-Lex](<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689>). Always verify the current consolidated text and obtain qualified legal advice for a real deployment.

9. Conclusion

LEIS MEMORY is best described as an evidence-bounded orientation preservation and reconstruction protocol. Its credibility comes from bounded claims, explicit unknowns, reproducible tests, independent observers and a visible lineage. The scientific programme is therefore frozen as a disciplined experiment, while empirical generalisation remains open until the evidence is earned.

Status: operational core documented; empirical generalisation active; legal and public-brain claims remain bounded.